# MATP32

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## Lecture 1

**Remark 1.** In this course, we denote the Lebesgue measure of $A \subseteq \mathbb{R}^d$ as $|A|$. For integrals against the Lebesgue measure, $\mathrm{d}\lambda(x)$ will be written as $\mathrm{d}x$.

**Definition 2.** The space of *locally integrable functions* is defined as

$$L^1_{loc}(\mathbb{R}^d) \coloneqq \left\{ f : \mathbb{R}^d \to \mathbb{C} \mid f \text{ measurable}, \int_K |f(x)|\mathrm{d}x < \infty \forall K \subseteq \mathbb{R}^d \text{ compact} \right\}.$$

**Definition 3.** For $f \in L^1_{loc}(\mathbb{R}^d)$ and $x \in \mathbb{R}^d$, define

$$(M_C f)(x) \coloneqq \sup_{r>0} \frac{1}{|B_r(x)|} \int_{B_r(x)} |f(y)|\mathrm{d}y$$

and

$$(Mf)(x) \coloneqq \sup_{\substack{B \ni x \\ B \text{ open ball}}} \frac{1}{|B|} \int_B |f(y)|\mathrm{d}y$$

as functions from $\mathbb{R}^d$ to $[0, \infty]$. These are called the *centered maximal function* and *maximal function* respectively.

**Proposition 4.** *For $f \in L^1_{loc}(\mathbb{R}^d)$, $x \in \mathbb{R}^d$, $M_C f(x) \leq Mf(x) \leq 2^d M_C f(x)$.*

*Proof.* The first inequality follows immediately from the fact that the set of open balls centered at $x$ is a subset of the set of open balls containing $x$. For the second inequality, note that if $x \in B$ for some open ball $B$ with radius $r$, then $B \subseteq B_{2r}(x)$ as all points of $B$ are at most $2r$ away from $x$. Thus,

$$\frac{1}{|B|} \int_B |f(y)|\mathrm{d}y \leq \frac{1}{|B|} \int_{B_{2r}(x)} |f(y)|\mathrm{d}y = \frac{2^d}{|B_{2r}(x)|} \int_{B_{2r}(x)} |f(y)|\mathrm{d}y,$$

and the corresponding inequality must hold for the suprema. □

**Remark 5.** Clearly, $\|Mf\|_\infty \leq \|f\|_\infty$, so $M$ can be seen as an operator from $L^\infty(\mathbb{R}^d)$ to $L^\infty(\mathbb{R}^d)$. For $L^1$, however, the analogous result is false. In fact, we have that if $f \in L^1(\mathbb{R})$ is non–trivial (i.e. $|\{x \mid |f(x)| > 0\}| > 0$), then

$Mf \notin L^1(\mathbb{R}^d)$. To show this, note that by our assumption, there must exist some $R > 0$ with $\int_{B_R(0)} |f(y)|\mathrm{d}y > 0$. Then, for $|x| > R$, we have that

$$\begin{aligned}\frac{1}{|B_{R+|x|}(0)|}\int_{B_{R+|x|}(0)} |f(y)|\mathrm{d}y &\geq \frac{1}{|B_{R+|x|}(0)|}\int_{B_R(0)} |f(y)|\mathrm{d}y \\ &\geq \frac{1}{|B_{2|x|}(0)|}\int_{B_R(0)} |f(y)|\mathrm{d}y \\ &\geq \underbrace{\frac{1}{x^d}}_{\text{too big to be integrable}} \underbrace{\frac{1}{|B_2(0)|}\int_{B_R(0)} |f(y)|\mathrm{d}y}_{\text{non–zero constant}},\end{aligned}$$

**Lemma 6** (Vitali covering lemma)**.** *Let $\{B_\ell\}_{\ell=1}^N$ be a finite collection of open balls in $\mathbb{R}^d$. Then, there is a finite subcollection $\{B_{\ell_j}\}_{j=1}^M$ which is pairwise disjoint and satisfies*

$$\left|\bigcup_{\ell=1}^N B_\ell\right| \leq 3^d \left|\bigcup_{\ell=1}^M B_{\ell_j}\right|.$$

*Proof.* Without loss of generality, the balls are ordered by size, i.e. $|B_1| \geq |B_2| \geq \cdots \geq |B_N|$. Define the subcollection recursively by $\ell_1 = 1$ and

$$\ell_{j+1} = \min\left\{k \in \{\ell_j, \ldots, N\} \mid B_k \cap \left(\cup_{i=1}^j B_{\ell_j}\right) = \varnothing\right\}$$

until that set is empty. In other words, we pick the largest ball, and keep picking the largest unpicked ball which is disjoint from the ones we've picked so far. Clearly, our chosen subcollection is pairwise disjoint. Now, it is enough to show that that

$$\bigcup_{\ell=1}^N B_\ell \subseteq \bigcup_{\ell=1}^M \underbrace{3B_{\ell_j}}_{\text{triple radius}}.$$

To this end, note that by our algorithm, all balls on the left hand side are either chosen, or intersect non–trivially with a larger, chosen ball. Moreover, it is a geometric fact that if two balls intersect non–trivially, then tripling the radius of the larger will make it fully contain the smaller (this can be shown using e.g. the triangle inequality). Taking the Lebesgue measures on both sides and factoring out the 3 gives the result. □

**Theorem 7.** *For $f \in L^1(\mathbb{R}^d)$ and $a > 0$, we have that*

$$|\{x \in \mathbb{R}^n \mid Mf(x) > a\}| \leq \frac{3^d}{a}\int |f(y)|\mathrm{d}y.$$

*Proof.* Denote the set in the left-hand side of the claim by $E_a$. Without loss of generality, $|E_a| > 0$. By the regularity of the Lebesgue measure, $|E_a| = \sup_{\substack{K \subseteq E_a \\ K \text{ compact}}} |K|$, so it is enough to show the inequality for all compact subsets

of $E_a$. Fix such a $K$, which is without loss of generality non–empty. Then, for each $x \in K$ let $B_x$ be an open ball containing $x$ such that

$$\frac{1}{|B_x|}\int_{B_x} |f(y)|\mathrm{d}y > a$$

($B_x$ exists because $x \in K \subseteq E_a$). Then, $K \subseteq \cup_{x\in K} B_x$, so by the compactness of $K$ there must exist a finite subcover $x_1, \ldots, x_N$ with $K \subseteq \cup_{i=1}^N B_{x_i}$. From these, use Lemma 6 to extract a subcollection $B_1', \ldots, B_M'$ with

$$|K| \leq \left|\bigcup_{i=1}^N B_{x_i}\right| \leq 3^d \left|\bigcup_{i=1}^M B_i'\right| = 3^d \sum_{i=1}^M |B_i'| \leq 3^d \sum_{i=1}^M \frac{1}{a}\int_{B_i'} |f(y)|\mathrm{d}y \leq \frac{3^d}{a}\int |f(y)|\mathrm{d}y.$$

□

# Lecture 2

**Definition 8.** Let $(X, \mathscr{A}, \mu)$ be a measure space, $0 < p \leq q \leq \infty$. Then, define $L^p(X) + L^q(X)$ as $\{f + g \mid f \in L^p(X), g \in L^q(X)\}$.

**Lemma 9.** *Let $(X, \mathscr{A}, \mu)$ be a measure space, $0 < r < p < q \leq \infty$. Then, $L^p(X) \subseteq L^r(X) + L^q(X)$.*

*Proof.* Let $f \in L^p(X)$. For $a > 0$ write

$$f = \underbrace{f \cdot \chi_{\{x||f(x)|>a\}}}_{f^a} + \underbrace{f \cdot \chi_{\{x||f(x)|\leq a\}}}_{f_a}.$$

We now claim that $f^a \in L^r(X)$ and $f_a \in L^q(X)$. Indeed,

$$\int |f^a|^r \mathrm{d}\mu = \int_{\{|f|>a\}} |f|^r \mathrm{d}\mu = \int_{\{|f|>a\}} |f|^p \cdot \underbrace{|f|^{\overbrace{r-p}^{<0}}}_{\leq a^{r-p}} \mathrm{d}\mu \leq \|f\|_p^p \cdot a^{r-p},$$

and

$$\int |f_a|^q \mathrm{d}\mu = \int_{\{|f|\leq a\}} |f|^q \mathrm{d}\mu = \int_{\{|f|\leq a\}} |f|^p \underbrace{|f|^{q-p}}_{\leq a^{q-p}} \mathrm{d}\mu \leq \|f\|_p^p \cdot a^{q-p}.$$

□

**Definition 10.** Let $U, V$ be vector spaces of functions from some set $X$ to $\mathbb{C}$. A function $T : U \to V$ is said to be a *sublinear operator* if $|(T(f+g))(x)| \leq |(Tf)(x)| + |(Tg)(x)|$ and $|(T(cf))(x)| = |c| \cdot |(Tf)(x)|$ for all $f, g \in U$, $c \in \mathbb{C}$, $x \in X$.

**Definition 11.** Let $(X, \mathscr{A}, \mu)$, $(Y, \mathscr{B}, \nu)$ be measure spaces, let $0 < p < \infty$, and let $T : L^p(X) \to L^p(Y)$. We say that $T$ is of *strong type* $(p, p)$ if there exists a constant $C_p$ with $\| Tf\|_{L^p(Y)} \leq C_p \|f\|_{L^p(X)}$ for all $f \in L^p(X)$. If instead there is a $C_p$ with

$$\nu\left(\{y \mid |Tf(y)| > a\}\right) \leq \frac{C_p}{a^p} \|f\|^p_{L^p(X)}$$

for all $f \in L^p(X)$, $a > 0$, we say that $T$ is of *weak type* $(p, p)$.

**Remark 12.** As the naming suggests, the strong type implies the weak type. Indeed, if $T$ is of strong type $(p, p)$, then

$$\begin{aligned}
\nu\left(\{y \mid |Tf(y)| > a\}\right) &= \frac{1}{a^p} \int_{\{|Tf|>a\}} a^p \mathrm{d}\nu \\
&\leq \frac{1}{a^p} \int_{\{|Tf|>a\}} |Tf|^p \mathrm{d}\nu \\
&\leq \frac{\|Tf\|^p_{L^p(Y)}}{a^p} \\
&\leq \frac{C_p}{a^p} \|f\|^p_{L^p(X)}.
\end{aligned}$$

**Theorem 13** (Marcinkiewicz interpolation theorem)**.** *Let $(X, \mathscr{A}, \mu)$, $(Y, \mathscr{B}, \nu)$ be $\sigma$–finite measure spaces, let $0 < q < r \leq \infty$ and let $T : L^q(X) + L^r(X) \to \{$functions $Y \to \mathbb{C}\}$ be a sublinear operator. If $T$ is of weak types $(q, q)$ and $(r, r)$ (unless $r = \infty$, in which case we want it to be of strong type $(r, r)$), then $T$ is of strong type $(p, p)$ for all $p \in (q, r)$.*

*Proof.* Let us first deal with the case $p < \infty$. First rewrite,

$$\begin{aligned}
\|Tf\|^p_{L^p(Y)} &= \int |Tf(y)|^p \mathrm{d}\nu(y) \\
&= \int \int_0^{|Tf(y)|} p\alpha^{p-1} \mathrm{d}\alpha \mathrm{d}\nu(y) \\
&= \int \int_0^{\infty} \chi_{\alpha \leq |Tf(y)|}(\alpha, y) p\alpha^{p-1} \mathrm{d}\alpha \mathrm{d}\nu(y) \\
&= \int_0^{\infty} p\alpha^{p-1} \int_{\{y \mid |Tf(y)|>\alpha\}} 1 \mathrm{d}\nu(y) \mathrm{d}\alpha \\
&= \int_0^{\infty} p\alpha^{p-1} \nu\left(\{y \mid |Tf(y)| > \alpha\}\right) \mathrm{d}\alpha.
\end{aligned}$$

Now, for $\alpha > 0$ decompose $f = \underbrace{f\chi_{|f|>\alpha}}_{f^\alpha} + \underbrace{f\chi_{|f|\leq\alpha}}_{f_\alpha}$. By sublinearity, we have that for all $y \in Y$, $|Tf(y)| \leq |Tf_\alpha(y)| + |Tf^\alpha(y)|$. Therefore,

$$\{y \in Y \mid |Tf(y)| > \alpha\} \subseteq \{y \in Y \mid |Tf^\alpha(y)| > \alpha/2\} \cup \{y \in Y \mid |Tf_\alpha(y)| > \alpha/2\},$$

which means that

$$\nu(\{|Tf(y)| > \alpha\}) \leq \nu(\{|Tf^{\alpha}(y)| > \alpha/2\}) + \nu(\{|Tf_{\alpha}(y)| > \alpha/2\})$$

and

$$\|Tf\|_{L^p(Y)}^p \leq \underbrace{\int_0^\infty p\alpha^{p-1}\nu\left(\{y \mid |Tf^{\alpha}(y)| > \alpha/2\}\right)\mathrm{d}\alpha}_{A} + \underbrace{\int_0^\infty p\alpha^{p-1}\nu\left(\{y \mid |Tf_{\alpha}(y)| > \alpha/2\}\right)\mathrm{d}\alpha}_{B}.$$

Because $T$ was assumed to be of weak type $(q, q)$, we have that

$$\begin{aligned}
A &\leq \int_0^\infty p\alpha^{p-1}\frac{C_q}{(\alpha/2)^q}\left(\int |f^{\alpha}|^q \mathrm{d}\mu\right)\mathrm{d}\alpha \\
&= 2^q pC_q \int_0^\infty \alpha^{p-q-1}\int_{\{|f|>\alpha\}} |f|^q \mathrm{d}\mu\mathrm{d}\alpha \\
&= 2^q pC_q \int |f(x)|^q \int_0^{|f(x)|} \alpha^{p-q-1}\mathrm{d}\alpha\mathrm{d}\mu(x) \\
&= \frac{2^q pC_q}{p-q}\int |f|^q |f|^{p-q}\mathrm{d}\mu \\
&= \frac{2^q pC_q}{p-q}\|f\|_{L^p(X)}^p.
\end{aligned}$$

Likewise,

$$\begin{aligned}
B &\leq \int_0^\infty p\alpha^{p-1}\frac{C_r}{(\alpha/2)^r}\left(\int |f_{\alpha}|^r \mathrm{d}\mu\right)\mathrm{d}\alpha \\
&= 2^r pC_r \int_0^\infty \alpha^{p-r-1}\int_{\{|f|\leq\alpha\}} |f|^r \mathrm{d}\mu\mathrm{d}\alpha \\
&= 2^r pC_r \int |f(x)|^r \int_{|f(x)|}^\infty \alpha^{p-r-1}\mathrm{d}\alpha\mathrm{d}\mu(x) \\
&= \frac{2^r pC_r}{r-p}\int |f|^r |f|^{p-r}\mathrm{d}\alpha\mathrm{d}\mu \\
&= \frac{2^r pC_r}{r-p}\|f\|_{L^p(X)}^p
\end{aligned}$$

Combining these gives that

$$\|Tf\|_{L^p(Y)}^p \leq p\left(\frac{2^q C_q}{p-q} + \frac{2^r C_r}{r-p}\right)\|f\|_{L^p(X)}^p,$$

and taking $p$–th roots gives the strong type constant.
In the other case, i.e. $r = \infty$, the assumption is that $\|Tf\|_{L^\infty(Y)} \leq C_\infty \|f\|_{L^\infty(X)}$.
Decompose $f$ as $f = \tilde{f}^\alpha + \tilde{f}_\alpha$ where $\tilde{f}^\alpha = f\chi_{|f|>\alpha/(2C_\infty)}$ and $\tilde{f}_\alpha = f\chi_{|f|\leq\alpha/(2C_\infty)}$.
Like in the first case, we have by sublinearity that,

$$|Tf(x)| \leq |T\tilde{f}^\alpha(x)| + |T\tilde{f}_\alpha(x)|$$

and

$$\nu\left(\{y \mid |Tf(y)| > \alpha\}\right) \leq \nu\left(\left\{y \mid |T\tilde{f}^\alpha(y)| > \alpha/2\right\}\right) + \nu\left(\left\{y \mid |T\tilde{f}_\alpha(y)| > \alpha/2\right\}\right).$$

The first term can be bounded like $A$ in the first case, while the second term is zero by the definition of $\tilde{f}_\alpha$. Indeed,

$$\nu\left(\left\{y \mid |T\tilde{f}_\alpha(y)| > \alpha/2\right\}\right) \leq \nu\left(\left\{y \mid C_\infty|\tilde{f}_\alpha(y)| > \alpha/2\right\}\right) = \nu\left(\underbrace{\left\{y \mid |\tilde{f}_\alpha(y)| > \alpha/(2C_\infty)\right\}}_{=\varnothing}\right).$$

□

**Corollary 14.** *For any $1 < p < \infty$, $M$ (i.e. the maximal function operator) is of strong type $(p, p)$.*

*Proof.* Theorem 7 and Remark 5 give the edge bounds for Theorem 13. □

# Lecture 3

**Theorem 15** (Lebesgue Differentiation theorem)**.** *Let $f \in L^1_{loc}(\mathbb{R}^d)$, then*

$$\frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy \to f(x)$$

*as $r \to 0$ for almost every $x \in \mathbb{R}^d$.*

*Proof.* We begin by letting by defining the operator $R$ on $L^1_{loc}(\mathbb{R}^d)$

$$R(f)(x) = f_r(x) := \frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy.$$

We then define the *ocsillation* which is defined on $L^1(\mathbb{R}^d)$,

$$\mathrm{osc}(f) = |\limsup_{r\to 0} f_r(x) - \liminf_{r\to 0} f_r(x)|.$$

It is easy to see that osc(f)(x) is less or equal to $2M_c f(x)$ and therefore also $2Mf(x)$. Additionnaly it is easy to see that if $f$ is continuous than the limit

$\lim_{r\to 0} f_r(x)$ exists and therefore the oscillation is zero. Now we note that if $f, h \in L^1(\mathbb{R}^d)$ and $h$ continuous where $h_r(x) \to c(x)$ then

$$2M(f+h)(x) \geq \operatorname{osc}(f+h) = |\limsup_{r\to 0}(f_r + h_r) - \liminf_{r\to 0}(f_r + h_r)| =$$
$$= |\limsup_{r\to 0} \frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)+h(y)dy - \liminf_{r\to 0}\frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)+h(y)dy| =$$
$$= |\limsup_{r\to 0} \frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy + c(x) - \liminf_{r\to 0}\frac{1}{|B_r(x)|}\int_{B_r(x)} f(y)dy - c(x)| =$$
$$= \operatorname{osc}(f)(x).$$

With this we aim at picking continuous $h$ so that $2M(f+h)$ becomes small forcing $\operatorname{osc}(f)(x)$ to zero. So pick arbirtary $\varepsilon > 0$ and continuous $h$ such that $||f+h||_1 \leq \varepsilon$. Now as the maximal function is of weak type $(1,1)$ we have that, for any $a > 0$,

$$\nu(\{y||M(f+h)| > a\}) \leq \frac{C_p}{a^p}||f+h||_1 \leq \frac{C_p\varepsilon}{a^p}.$$

Since $\varepsilon$ is arbitrary we get that for any $\delta > 0$ there exists an $\varepsilon > 0$ such that

$$\nu(\{\operatorname{osc}(f) > \delta\}) \leq \varepsilon$$

. This gives that osc(f) is zero almost everywhere. What we have proven is that the limit

$$\lim_{r\to 0} f_r(x)$$

exists almost everywhere and what remains is to show that it is equal to $f(x)$ almost everywhere. TODO □

**Corollary 16.** *For almost every $x \in \mathbb{R}^d$ we have that, for $f \in L^1_{loc}(\mathbb{R}^d)$, $|f(x)| \leq Mf(x)$. This gives that for each $p \in (1,\infty]$ there exists a $C_{p,d} > 0$ such that $||f||_p \leq ||Mf||_p \leq C_{p,d}||f||_p$.*

**Definition 17.** A family $(k_t)_{t>0}$ of functions in $L^1(\mathbb{R}^d)$ is called an approximate identity if
(i) $\int_{\mathbb{R}^d} k_t(x)dx = 1 \quad \forall t > 0$;
(ii) $\exists C > 0 : ||k_t||_1 < C \quad \forall t > 0$;
(iii) $\forall \delta > 0 \quad \int_{|x|>\delta} |k_t(x)|dx \to 0$ as $t \to 0$.

**Example 18.** Let $k \in L^1(\mathbb{R}^d)$ with $\int_{\mathbb{R}^d} k(x)dx = 1$. Then we can define $k_t(x) := t^{-d}k(t^{-1}x)$ and this will be an approximate identity.

**Theorem 19.** *Let $(k_t)_{t>0}$ be an approximate identity, then*
*(i) if $f \in L^p(\mathbb{R}^d), 1 \leq p < \infty$, then*

$$||k_t * f - f||_{L^p} \to 0 \text{ as } t \to 0;$$

*(ii) if $f \in C_0(\mathbb{R}^d)$, then*

$$k_t * f \to f \text{ uniformly as } t \to 0.$$

*Proof.* For (i) we let $f \in L^p(\mathbb{R}^d), 1 \leq p < \infty$. Now we write

$$k_t * f - f = \int_{\mathbb{R}^d} k_t(x)(f(x-y) - f(x))dy$$

and note that if $f$ was continuous we could split the integral into one of a sphere and one of an anti-sphere defined by some delta. $(f(x-y) - f(x))$ would be small in the sphere and $k_t(y)$ would be small in the anti-sphere and we would be done. However, of course $f$ is not necessarily continuous and we cannot do this. Except we can, as we shall see.

**Lemma 20.** *For $y \in \mathbb{R}^d$ and $1 \leq p < \infty$ let $T_y : L^p(\mathbb{R}^d) \to L^p(\mathbb{R}^d)$ send $f(x)$ to $f(x-y)$. Then for $f \in L^p(\mathbb{R}^d)$ we get*

$$||T_y f - f||_p \to 0 \text{ as } y \to 0.$$

*Proof.* TODO □

Now we do

$$\begin{aligned} ||k_t * f - f||_p &\leq (\int_{\mathbb{R}^d} (\int_{\mathbb{R}^d} |k_t(x)||f(x-y) - f(x)|dy)^p dx)^{1/p} \leq \\ &\leq \int_{\mathbb{R}^d} |k_t(y)| (\int_{\mathbb{R}^d} |f(x-y) - f(x)|^p dx)^{1/p} dy \\ &= \int_{\mathbb{R}^d} |k_t(y)| \cdot ||T_y f - f||_p dy \end{aligned}$$

The last inequality follows from Minkowski's inequality (apparently?). Now we can do the split up we wanted to do in the continuous case.

(ii) TODO (as in the Fourier analysis course apparently) □

We will now generelise the Lebesgue Differentiation Theorem, starting with an example.

**Example 21.** Let $k = V_d^{-1} \chi_{B_1(0)}$ where $V_d^{-1} = |B_1(0)|$ and to make an approximate identity define

$$k_t = t^{-d} k(t^{-1}x)$$

then we have that

$$\begin{aligned} M_c f(x) &= \sup_{t>0} \frac{1}{|B_t(x)|} \int_{B_t(x)} |f(y)|dy = \\ &= \sup_{t>0} \frac{1}{t^d V_d} \int_{B_t(0)} |f(x-y)|dy = \\ &= \sup_{t>0} t^{-d} V_d^{-1} \int_{\mathbb{R}^d} \chi_{B_1(0)}(t^{-1}y) \cdot |f(x-y)|dy = \\ &= \sup_{t>0} k_t * |f|. \end{aligned}$$

In other words the centered maximal function arises from convolution with a certain approximal distribution. This can be generalized.

**Theorem 22.** *Let $g$ be a function from $[0,\infty)$ to $[0,\infty)$ which has at most finitely many discontinuities and is decreasing. Consider $k : \mathbb{R}^d \to \mathbb{R}$ given by $k(x) = g(|x|)$. If $k \in L^1(\mathbb{R}^d)$ then $\forall f \in L^1_{loc}(\mathbb{R}^d)$ one has*

$$\sup_{t>0}(|f| * k_t)(x) \leq ||k||_1 M_c f$$

*for almost every $x \in \mathbb{R}^d$.*

No proof

**Corollary 23.** *Let $g$ be as in the theorem above. Let further $k \in L^1(\mathbb{R}^d)$, $\int_{\mathbb{R}^d} k dx = 1$ and $|k(x)| \leq g(|x|)$. Then for $f \in L^p(\mathbb{R}^d)$, $p \in (1,\infty)$ we have*

$$k_t * f \to f(x) \text{ as } t \to 0$$

*for almost every $x \in \mathbb{R}^d$.*
*How does this follow? TODO*

We will now deal with the Schwartz class and it's dual.

**Definition 24.** The Schwartz Class denoted $\mathcal{S}(\mathbb{R}^d)$ encompasses all functions $f \in C^\infty(\mathbb{R}^d)$ satisfying

$$p_{a,b}(f) := \sup_{x \in \mathbb{R}^d} |x^a \partial^b f(x)| < \infty \quad \forall a, b \in \mathbb{N}_0^d.$$

Here $a$ and $b$ are multi-indices and $\partial^b = \partial_1^{b_1} \cdots \partial_d^{b_d}$, $x^a = x_1^{a_1} \cdots x_d^{a_d}$. We have in this space a metric induced by all the $p_{a,b}$ norms defined as

$$d(f,g) = \sum_{j=1}^{\infty} 2^{-j} \frac{p_j(f-g)}{1 + p_j(f-g)}$$

where $(p_j)$ is an enumeration of all the $p_{a,b}$'s. See that we converge in this metric if and only if we converge in each $p_{a,b}$ norm.

This is a space with particularly nice functions. For example the Fourier transform maps the Schwartz space to itself.

**Definition 25.** A linear map $u : \mathcal{S}(\mathbb{R}^d) \to \mathbb{C}$ is called bounded if

$$u(\varphi_k \to 0 \text{ as } k \to 0$$

whenever $\varphi_k \to 0$ in $d$.

**Definition 26.** The space of tempered distributions is defined as the dual of $\mathcal{S}(\mathbb{R}^\lceil)$ which is the space of bounded linear maps $\mathcal{S}(\mathbb{R}^d) \to \mathbb{C}$. it is denoted

$$\mathcal{S}(\mathbb{R}^d)'.$$

# Lecture 4

The space of tempered distributions is very big. For example note that for ant $f \in L^1(\mathbb{R}^d)$ we may for $\varphi \in \mathcal{S}(\mathbb{R}^d)$ identify

$$\varphi \mapsto < f, \varphi >= \int_{\mathbb{R}^d} f \cdot \overline{\varphi} dx$$

which defines a tempered distribution. Connecting tempered distributions to functions on $\mathbb{R}^d$ this way will be quite central so we will, for $u$ a tempered distribution, write $< u, \varphi >$ for $u(\varphi)$.

**Example 27.** Some examples of distributions:
1). The dirac delta function $\delta_0$, maps $\varphi$ to $\varphi(0)$.
2). Any $L^p$ function gives a distribution. 3). Define $p.v.(\frac{1}{x}) : \mathcal{S}(\mathbb{R}^d) \to \mathbb{C}$ by

$$< p.v.(\frac{1}{x}), \varphi >= \lim_{\varepsilon \to 0^+} \int_{\varepsilon \leq |x| \leq 1} \frac{\varphi(x)}{x} dx + \int_{|x|>1} \frac{\varphi(x)}{x} dx.$$

This is a tempered distribution (p.v. stand for principal valued).

**Definition 28.** We will define what is means to differentiate a distribution. We do this through the connection to normal functions in the inner product. First recall that with partial integration we have for $\varphi, \psi \in \mathcal{S}(\mathbb{R}^d)$ that

$$\int_{\mathbb{R}^d} \partial^a \varphi(x) \psi(x) dx = \int_{\mathbb{R}^d} \varphi(x) \partial^a \psi(x) dx.$$

Now for $u$ a tempered distribution define $\partial^b u$ by defining haw it acts on $\mathcal{S}(\mathbb{R}^d)$,

$$< \partial^b u, \varphi >=< u, \partial^b \varphi > .$$

Now we define $\partial^a u$ by specifying how it acts on the Schwartz space

$$\partial^a u(\varphi) =< \partial^a u \, \varphi >:=< u, \partial^a \varphi > .$$

**Definition 29.** Now for convolution. Recall that for $\varphi, \psi, \eta \in \mathcal{S}(\mathbb{R}^d)$ and $\tilde{\eta}(x) = \eta(-x)$ we have

$$< \eta * \varphi, \psi >=< \varphi, \psi * \tilde{\eta} >$$

so we define $u * \eta$ by

$$< u * \eta, \varphi >:=< u, \varphi * \tilde{\eta} > .$$

**Theorem 30.** *If $u \in \mathcal{S}(\mathbb{R}^d), \varphi \in \mathcal{S}(\mathbb{R}^d)$ then $u * \varphi$ agrees with a $C^\infty$ function.*

**Definition 31.** Define the fourier transform $\mathcal{F}$ as

$$\mathcal{F}(\varphi)(\xi) := \int_{\mathbb{R}^d} \varphi(x) e^{-2\pi i x \xi} dx$$
$$\mathcal{F}^{-1}(\varphi)(\xi) := \int_{\mathbb{R}^d} \varphi(x) e^{2\pi i x \xi} dx$$

and note that it sends $\mathcal{S}(\mathbb{R}^d)$ to itself and that as a functional on $\mathcal{S}(\mathbb{R}^d)$ is satifies

$$\begin{aligned}&\mathcal{F}\circ\mathcal{F}^{-1}=id;\\&\mathcal{F}^{-1}\circ\mathcal{F}=id;\\&<\varphi,\psi>=<\mathcal{F}(\varphi),\mathcal{F}(\psi)>.\end{aligned}$$

Thus we define $\mathcal{F}(u)$ as

$$<\mathcal{F}(u),\varphi>:=<u,\mathcal{F}^{-1}(\varphi)>.$$

**Proposition 32.** *Let $k\in L^1(\mathbb{R}^d)$ with $\int_{\mathbb{R}^d}k dx=1$ and let*

$$k_t:=t^{-d}k(t^{-1}x),$$

*and approximate identity. then $k_t\to\delta_0$ in $\mathcal{S}(\mathbb{R}^d)'$ as $t\to 0$. In other words*

$$<k_t,\varphi>\to<\delta_0,\varphi>=\varphi(0),\quad\forall\varphi\in\mathcal{S}(\mathbb{R}^d).$$

*Proof.*

$$\begin{aligned}<k_t,\varphi>&=\int_{\mathbb{R}^d}t^{-d}k(t^{-1}x)\varphi(x)dx=\\&=\int_{\mathbb{R}^d}k(y)\varphi(ty)dy\to\int_{\mathbb{R}^d}k(y)\varphi(0)dy=\varphi(0)\end{aligned}$$

□

$\delta_0$ is the identity in convolution so this explains why approximate identities are called approximate identities.

**Definition 33.** If $X$ and $Y$ are two vector spaces of $\mathbb{R}^d$-functions that are closed under translation and $T$ is an operator between them we say that $T$ is translation invariant if it commutes with translations, i.e.

$$T\circ\tau_h=\tau_h\circ T.$$

**Theorem 34.** *Suppose that $T:L^2(\mathbb{R}^d)\to L^2(\mathbb{R}^d)$ is a lenear, bounded and translation invariant operator. Then there exists $u\in\mathcal{S}(\mathbb{R}^d)'$ such that*

$$\begin{aligned}&Tf\text{ agrees with }u*f,\quad\forall f\in\mathcal{S}(\mathbb{R}^d);\\&\mathcal{F}(u)\text{ agrees with an }L^\infty\text{-function.}\end{aligned}$$

*Proof.* No proof (for some reason I wrote 'duandikoetxea' in my notes) □

**Theorem 35** (Calderon-Zygmund operator of convolution type)**.** *Suppose that $u\in\mathcal{S}(\mathbb{R}^d)'$ is such that*

$$\begin{aligned}&\cdot\,\mathcal{F}(u)\text{ agrees with an }L^\infty\text{-function on }\mathbb{R}^d;\\&\cdot\,u\text{ agrees on }\mathbb{R}^d\backslash\{0\}\text{ with a }L^1_{loc}(\mathbb{R}^d\backslash\{0\})\text{ function }k\text{ satisfying}\\&\sup_{y\in\mathbb{R}^d}\int_{|x|>2|y|}|k(x-y)-k(x)|dx<\infty.\end{aligned}$$

*Then the operator $T : \mathcal{S}(\mathbb{R}^d) \to \mathcal{S}(\mathbb{R}^d)$ sending $\varphi$ to that which agrees with $u * \varphi$ can be extended to a bounded linear operator $T : L^p(\mathbb{R}^d) \to L^p(\mathbb{R}^d)$ for each $1 < p < \infty$.*

Before we can prove this we need some buildup.

**Definition 36.** A dyadic cube is a subset of $\mathbb{R}^d$ of the form

$$2^n[a_1, a_1 + 1) \times \cdots \times 2^n[a_d, a_d + 1)$$

where $a_i, n \in \mathbb{Z}$. Picture $\mathbb{R}^d$ space and divide it into a grid of cubes with side-length $2^n$. These cubes are dyadic cubes and the set of dyadic cubes is the collection of cubes from all such grids. Note also by the way we defined the cubes with the half-open intervals that for any two dyadic cubes they are either disjoint or one of them is contained in the other.

**Theorem 37** (Calderon-Zygmund Decomposition)**.** *Let $f \in L^1(\mathbb{R}^d)$ and let $a > 0$. Then there exists collection $\mathcal{Q}$ of disjoint dyadic cubes such that, where $F = \mathbb{R}^d \backslash \bigcup_{Q \in \mathcal{Q}} Q$, we have*

$$(i) \quad |f| \leq a \text{ almost everywhere on } F;$$
$$(ii) \quad a < \frac{1}{|Q|} \int_Q |f| dx \leq 2^d a \quad \forall Q \in \mathcal{Q}.$$

# Lecture 5

*Proof.* Since $f$ is $L^1$ we can choose an $N > 0$ such that

$$\frac{1}{|Q|} \int_Q |f| dx \leq a$$

for any dyadic cube $Q$ with sidelenght $2^{dN}$. We now consider all dyadic cubes with sidelenght $2^{dN}$. Let $Q$ be such a dyadic cube and divide it into $2^d$ dyadic cubes with half the sidelenght. Take one such cube $Q'$, for this there are two cases

$$\textbf{Case 1: } \frac{1}{|Q'|} \int_{Q'} |f| dx > a$$
$$\textbf{Case 2: } \frac{1}{|Q'|} \int_{Q'} |f| dx \leq a$$

In the first case we add the $Q'$ into what will become the desired collection, $\mathcal{Q}$. This is justified as

$$\frac{1}{|Q'|} \int_{Q'} |f| dx \leq \frac{2^d}{|Q|} \int_Q |f| dx \leq 2^d a$$

and note that it would also be justified if $Q$ just satisfied **Case 2**. In the second case we split up $Q'$ just like $Q$ and look at the cases again. This process is

continued forever and defines a collection $\mathcal{Q}$ that satisfies the second property of the theorem. For the first, let $x \in F := \mathbb{R}^d \setminus \bigcup_{Q\in\mathcal{Q}} Q$. Then there is an infinite series of dyadic cubes $(Q_i)$ satisfying **Case 2:** such that $x = \bigcap Q_i$. After applying a variant of Lebesgue's differentiation theorem to $|f|$ we find that $|f| < a$ for almost every $x \in F$. □

**Remark 38.** If $f \in L^1(\mathbb{R}^d)$ and $a > 0$ then we can construct $\mathcal{Q}$ and $F$ as in the previous theorem. Then we can define a function $g$ as

$$g = \begin{cases} f \text{ on } F \\ \frac{1}{Q}\int_Q f(y)dy \text{ on } Q \end{cases}$$

and $b$ as $b := f - g$. Then we have the following properties:

$$\begin{aligned} &(i) \quad f = g + b; \\ &(ii) \quad |g| \le 2^d a \text{ almost everywhere}; \\ &(iii) \quad \operatorname{supp}(b) \subseteq \bigcup_{Q\in\mathcal{Q}} Q, \quad \int_Q b(x)dx = 0; \\ &(iv) \quad \frac{1}{|Q|}\int_Q |b(x)|dx \le 2^{d+1}a. \end{aligned}$$

This description of $f$ will be used in the proof below.

*Proof of Calderon-Zygmund operator of convolution type Theorem.* The overall strategy of the proof is to first prove that the statement holds for $p = 2$. Then prove that the operator $T$ is of weak type $(1, 1)$ and use Marcinkiewicz interpolation theorem to conclude that the statement holds for $p \in (1, 2)$. Then the statement for $p \in (2, \infty)$ is proven with a duality argument.
We begin with $p = 2$ TODO
For $p = 1$ we let $f \in \mathcal{S}(\mathbb{R}^d)$ and use the Calderon-Zygmund decomposition theorem to choose collection $\mathcal{Q}$ of dyadic cubes and to split $f = g + b$. First we need to show that $g$ and $b$ lie in $L^2$ and here it is enough to show the belonging for one of them as $f$ lies in $L^2$.

$$\begin{aligned} \int_{\mathbb{R}^d} |g|^2 dx &\le 2^d a \int_{R^d} |g|dx = 2^d a\Big(\int_F |g|dx + \int_{\bigcup_{Q\in\mathcal{Q}} Q} |g|dx\Big) = \\ &= 2^d a\Big(\int_F |f|dx + \int_{\bigcup_{Q\in\mathcal{Q}} Q} |\frac{1}{|Q|}\int_Q f(y)dy|dx\Big) \le \\ &\le 2^d a(||f||_1 + \sum_{\bigcup_{Q\in\mathcal{Q}} Q} \int_Q |f(y)|dy) \le \\ &\le 2^{d+1} a||f||_1 \end{aligned}$$

so $g$ is $L^2$. We want to show that $\{|Tf(x)| > a\} \le \frac{C}{a}||f||_1$, so, using that $g$ and $b$ lie in $L^2$ meaning that $Tf = Tg + Tb$, we do

$$|\{|Tf(x)| > a\}| = |\{|Tg(x) + Tb(x)| > a\}| \le |\{|Tg(x)| > \frac{a}{2}\}| + |\{|Tb(x)| > \frac{a}{2}\}|$$

and see that we need to find $C_1$ and $C_2$ such that

$$|\{|Tg(x)| > \frac{a}{2}\}| \leq \frac{C_1}{a}||f||_1$$
$$|\{|Tb(x)| > \frac{a}{2}\}| \leq \frac{C_2}{a}||f||_1.$$

We start with the $g$ term.

$$\begin{aligned}|\{|Tg(x)| > \frac{a}{2}\}| &\leq \frac{1}{(\frac{a}{2})^2}\int_{\mathbb{R}^d}|Tg(x)|^2dx = \\ &= \frac{4}{a^2}||u*g||_2^2 = \frac{4}{a^2}||\mathcal{F}(u*g)||_2^2 = \\ &= \frac{4}{a^2}||\mathcal{F}(u)\cdot\mathcal{F}(g)||_2^2 \leq \frac{4}{a^2}||\mathcal{F}(u)||_\infty\cdot||(g)||_2^2 \leq \\ &\leq \frac{4\cdot 2^{d+1}||\mathcal{F}(u)||_\infty}{a}||f||_1.\end{aligned}$$

# Lecture 6

For the $b$ term we start by dividing it further, here $Q^*$ is the cube with the same center as $Q$ but with a sidelenght scaled by $2d^{1/2}$.

$$\begin{aligned}|\{|Tb(x)| > \frac{a}{2}\}| &\leq |\{x \in \bigcup_{Q\in\mathcal{Q}} Q^* : |Tb| > \frac{a}{2}\}| + |\{x \notin \bigcup_{Q\notin\mathcal{Q}} Q^* : |Tb| > \frac{a}{2}\}| \leq \\ &\leq |\bigcup_{Q\in\mathcal{Q}} Q^*| + |\{x \notin \bigcup_{Q\notin\mathcal{Q}} Q^* : |Tb| > \frac{a}{2}\}|\end{aligned}$$

The $|\bigcup_{Q\in\mathcal{Q}} Q^*|$ we know how to control from the Calderon-Zygmund decomposition theorem,

$$|\bigcup_{Q\in\mathcal{Q}} Q^*| = d^{d/2}2^d\sum_{Q\in\mathcal{Q}}|Q| \leq \frac{d^{d/2}2^d}{a}||f||_1.$$

For the second term we need to use the following results for which we omit the proofs.

**Claim:** *Let $Q$ be a cube in $\mathbb{R}^d$, if $b_Q \in L^\infty$ with $supp(b_Q) \subseteq Q$. Then for almost every $x \in \mathbb{R}^d\backslash Q^*$,*

$$Tb_Q(x) = \int_Q k(x-y)b_Q(y)dy.$$

**Claim:** *For the $b$ we are working with we have for $b_Q := \chi_Q b$,*

$$|T(\sum_{Q\in\mathcal{Q}} b_Q)| \leq \sum_{Q\in\mathcal{Q}}|Tb_Q|$$

*for almost every $x \in \mathbb{R}^d$.*

Second term,

$$
\begin{aligned}
&|\{x \notin \bigcup_{Q \notin \mathcal{Q}} Q^* : |Tb| > \frac{a}{2}\}| && \leq \\
\leq &|\{x \notin \bigcup_{Q \notin \mathcal{Q}} Q^* : |T(\sum_{Q \in \mathcal{Q}} b_Q)| > \frac{a}{2}\}| && \leq \\
\leq &|\{x \notin \bigcup_{Q \notin \mathcal{Q}} Q^* : \sum |Tb_Q| > \frac{a}{2}\}| && \leq \\
\leq &|\{x \notin \bigcup_{Q \notin \mathcal{Q}} Q^* : \sum |\int_Q k(x-y) b_Q(y) dy| > \frac{a}{2}\}| && \leq \\
\leq &\frac{2}{a} \int_{\mathbb{R}^d \setminus \bigcup Q^*} \sum |\int_Q k(x-y) b_Q(y) dy| dx && \leq \\
\leq &\frac{2}{a} \sum \int_{\mathbb{R}^d \setminus \bigcup Q^*} |\int_Q k(x-y) b_Q(y) dy| dx && \leq \\
&\textit{use that } b_Q \textit{ integrates over } Q \textit{ to zero, } C_Q \textit{ is the center of } Q \\
\leq &\frac{2}{a} \sum \int_{\mathbb{R}^d \setminus \bigcup Q^*} |\int_Q (k(x-y) - k(x-C_Q)) b_Q(y) dy| dx && \leq \\
\leq &\frac{2}{a} \sum \int_{\mathbb{R}^d \setminus \bigcup Q^*} \int_Q |(k(x-y) - k(x-C_Q)) b_Q(y)| dy dx && \leq \\
\leq &\frac{2}{a} \sum \int_Q |b_Q(y)| \int_{\mathbb{R}^d \setminus \bigcup Q^*} |k(x-y) - k(x-C_Q)| dx dy && \leq \\
\leq &\frac{2}{a} A \sum \int_Q |b_Q(y)| dy \leq 2^{d+2} A \sum |Q| \leq \frac{2^{d+2} M}{a} ||f||_1.
\end{aligned}
$$

With this we can add everything together and arrive at

$$
|\{|Tf| > a\}| \leq \frac{2^d (4 ||\mathcal{F}(u)||_\infty^2 + d^{d/2} + 4A)}{a} ||f||_1 = \frac{C}{a} ||f||_1
$$

for $f \in \mathcal{S}(\mathbb{R}^d)$. We now need to extend this to $f \in L^1$. For $f \in L^1(\mathbb{R}^d)$ choose $(f_n)$ in $\mathcal{S}(\mathbb{R}^d)$ with $||f - f_n||_1 \to 0$. Then $Tf_n$ is Cauchy in measure, meaning that $\forall \varepsilon > 0, \delta > 0$ there exists an $N \in \mathbb{N}$ such that

$$
|\{|Tf_n(x) - Tf_m(x)| > \varepsilon\}| \leq \delta
$$

for $n, m > \mathbb{N}$. Hence $Tf_n$ converges in measure to some $g$. Now choose another sequence $(f_n')$ just as $f_n$ and let $g'$ be what $Tf_n'$ converges to. As $Tf_n - Tf_n'$ converges to zero in measure $g$ must equal $g'$ and this is our definition for $Tf$. From this definition it follows that $T$ is of weak type $(1,1)$. Using Marcinkiewicz Interpolation Theorem we get for all $p \in (1,2)$ $T$ as a bounded operator on $L^p$ (We also get some bound on the operator norm which I do not understand).

Lastly let $p \in (2,\infty)$, $p' : \frac{1}{p} + \frac{1}{p'} = 1$ and $f \in \mathcal{S}(\mathbb{R}^d)$. Then

$$\begin{aligned} ||Tf||_p &= \sup_{g\in\mathcal{S}(\mathbb{R}^d):||g||_{p'}=1} |\int_{\mathbb{R}^d} Tf(x)\overline{g(x)}dx| = \sup|<u*f,\bar{g}>| = \\ &= \sup|<u,\bar{g}*\tilde{f}>| = \sup|<u*\tilde{\bar{g}},\tilde{f}>| = \\ &= \sup|\int_{\mathbb{R}^d} T\tilde{\bar{g}}(x)\tilde{f}(x)dx| \leq ||T||_{L^{p'}\to L^{p'}}||f||_p. \end{aligned}$$

We see that $||T||_{L^{p'}\to L^{p'}} = ||T||_{L^p\to L^p}$ and thus $T : L^p \to L^p$ is bounded for $p \in (1,\infty)$. □

Recall partial sums of the Fourier series.

$$S_N f(x) = \sum_{u=-N}^{N} \hat{f}(u)e^{2\pi iux}, \quad f : S_1 \to \mathbb{C}$$

If $f$ is $L^2$ we have that $S_N f$ converges to $f$ is the $L^2$-norm. But what about point-wise convergence?
Lennart Carleson (alive today) proved in 1966 that for $f \in L^2$ the Fourier series $S_N f$ converges to $f$ almost everywhere.
Richard Hunt generalized this result 1967 to hold for all $f \in L^p$ for $p \in (1,\infty)$.

Write $S^*f(x) = \sup_{k\in\mathbb{N}}|S_{2^k}f(x)|$. (And then we didn't do anything with this?)

A function:

$$S_\pi f(x) = \left(|S_1 f(x)|^2 + \sum_{k=1}^{\infty}|S_{2^k}f(x) - S_{2^{k-1}}f(x)|^2\right)^{1/2}.$$

In the 1920's the following inequality was showed for $f \in L^p$, $p \in (1,\infty)$,

$$A_p||f||_p \leq ||S_\pi f||_p \leq B_p||f||_p.$$

Now let $(a_k)$ be a sequence in $\ell^2$ and let $f$ be such that

$$\hat{f} = \begin{cases} a_k \text{ if } u = 2^k \\ 0 \text{ otherwise.} \end{cases}$$

Then $f = \sum_{u=0}^{\infty}\hat{f}(u)e^{2\pi iux} \in L^2$ and $||f||_2 = \left(\sum|a_k|^2\right)^{1/2}$. Now apparently we get something nice for $||S_\pi f||_p$ and therefore also for $||f||_p$.

# Lecture 7

We want to prove the series of inequalities concerning $S_\pi$. This function can be thought of as taking the Fourier series and cutting it into chunks that double in

length. However, Fourier series do not generally behave nicely when cut up so sharply in this manner, we will therefore first prove a version of this inequality for a function similar to $S_\pi$ but where the Fourier series is cut up "smoothly". For this let us pick $\eta \in C_c^\infty(\mathbb{R}^d)$ such that it is bounded by 1, equal to 1 in the closed unit disc, and zero outside the open disc of radius 2. We then define

$$\delta := \eta(\xi) - \eta(2\xi)$$

and note that the sum

$$\sum_{j=-\infty}^{\infty} \delta(2^{-j}\xi)$$

is telescoping and evaluates to 1 for $\xi \neq 0$. We write $\delta(2^{-j}\xi)$ as $\delta_j(\xi)$.

**Definition 39.** Let $\Delta_j f := \mathscr{F}^{-1}(\delta_j) * f = \check{\delta}_j * f$. Then define the Littlewood-Paley square function $S$, as

$$Sf := \left( \sum_{-\infty}^{\infty} |\Delta_j f|^2 \right)^{1/2}$$

This is our smooth version of $S_\pi$. Each $\Delta_j f$ can be understood as a smooth version of $S_{2^j} f - S_{2^{j-1}} f$. What we now seek to prove is then, that for any $p \in (1, \infty)$ there exits constants such that

$$A_{p,d,\eta} ||f||_p \leq ||Sf|| p \leq B_{p,d,\eta} ||f||_p.$$

**Lemma 40** (Khintchine's inequalities)**.** *Let $(\mathcal{O}, \mathscr{A}, \mathbb{P})$ be a probability space. We say that the independent random variables $T_n : \mathcal{O} \to \mathbb{R}$ are Rademacher if*

$$\mathbb{P}(\{w \in \mathcal{O} : T_n(w) = 1\}) = \mathbb{P}(\{w \in \mathcal{O} : T_n(w) = -1\}) = \frac{1}{2}.$$

*For each $p \in (0, \infty)$ there exists $A_p, B_p > 0$ such that for all $N \in \mathbb{N}$ and $a_1, \ldots, a_N \in \mathbb{C}$ we have*

$$A_p || \sum_{n=1}^{N} a_n T_n ||_p \leq \left( \sum_{n=1}^{N} |a_n|^2 \right)^{1/2} \leq B_p || \sum_{n=1}^{N} a_n T_n ||_p.$$

*Proof of the Littlewood-Paley square function inequalities.* Choose let $(\mathcal{O}, \mathscr{A}, \mathbb{P})$ be a probability space and choose Rademacher functions $\{V_j\}_{j \in \mathbb{N}}$. For $w \in \mathcal{O}$ and $N \in \mathbb{N}$ write

$$m_{w,N}(\xi) := \sum_{j=-N}^{N} V_j(w) \delta_j(\xi).$$

This is a finite sum Schwartz functions and thus a Schwartz function. Now let $T_{w,N} f := \check{m}_{m,N} * f$, then

$$T_{w,N} f = \sum_{j=-N}^{N} V_j(w) \check{\delta}_j * f = \sum_{j=-N}^{N} V_j \Delta_j f.$$

We shall prove that there exists an $A_{d,\eta} > 0$ such that

$$1) : ||m_{w,N}||_\infty \leq A_{d,\eta}$$
$$2) : \sup_{y\in\mathbb{R}^d} \int_{|x|>2|y|} |\check{m}_{w,N}(x-y) - \check{m}_{w,N}(x)|dx \leq A_{d,\eta}.$$

This would allow us to use the theorem on Calderon-Zygmund convolution type and prove that $T_{w,N}$ is bounded on $L^p(\mathbb{R}^d)$ for $p \in (1,\infty)$. Assuming we have this we do the following

$$\begin{aligned} C_{p,d}^p \int_{\mathbb{R}^d} |f(x)|^p dx &= \int_{\mathcal{O}} C_{p,d}^p \int_{\mathbb{R}^d} |f(x)|^p dx d\mathbb{P} \geq \\ &\geq \int_{\mathcal{O}} \int_{\mathbb{R}^d} |T_{w,N} f(x)|^p dx d\mathbb{P} = \\ &= \int_{\mathbb{R}^d} \int_{\mathcal{O}} |\sum_{j=-N}^{N} V_j \Delta_j f(x)|^p d\mathbb{P} dx \geq \\ &\geq B_p^{-p} \int_{\mathbb{R}^d} \left( \sum_{j=-N}^{N} |\Delta_j f(x)|^2 \right)^{p/2} dx \end{aligned}$$

which this gives us

$$||Sf||_p \leq B_p C_{p,d} ||f||_p.$$

This is one of the inequalities we want to prove. so we now need to prove 1) and 2). For 1) simply see

$$|m_{w,N}(\xi)| \leq \sum_{-N}^{N} |V_j(w)\delta_j(\xi)| \leq 6||\eta||_\infty.$$

For 2) note that for $g \in \mathcal{S}(\mathbb{R}^d)$, $a, b \in \mathbb{N}^d$ we have

$$(-2\pi i x)^b \partial^a \check{g}(x) = \int_{\mathbb{R}^d} \partial^b (P_a \cdot g)(\xi) e^{2\pi i x\xi} d\xi.$$

One can show that $|x|^M \leq C_{d,M} \sum_{|b|=M} |x^b|$ which gives

$$\begin{aligned} &(2\pi)^M |x|^b |\partial^a \check{\delta}_j(x)| \leq \int_{2^{j-1}\leq|\xi|\leq 2^{j+1}} |\partial^b (P_a \delta_j)(\xi)| d\xi \implies \\ &|\partial^a \check{\delta}_j(x)| \leq C_{d,M} |x|^{-M} \sum_{|b|=M} \int_{2^{j-1}\leq|\xi|\leq 2^{j+1}} |\partial^b (P_a \delta_j)(\xi)| d\xi \implies \\ &|\partial^a \check{m}_{w,N}| \leq B_{d,\eta,N} |x|^{-(|a|+d)} \quad \forall \xi \in \mathbb{R}^d \backslash \{0\}. \end{aligned}$$

Somehow this gives 2) (She was not more clear than this).
Now, to prove the other inequality we need first two lemmas.

**Lemma 41.** *Let $f, g \in \mathcal{S}(\mathbb{R}^d)$, $k \in \mathbb{Z}$. Then*

$$\int_{\mathbb{R}^d} \Delta_k f(x) \Delta_l g(x) dx = 0$$

*for $|k-l| > 1$.*

*Proof.*

$$\int_{\mathbb{R}^d} \Delta_k f(x) \Delta_l g(x) dx = \int_{\mathbb{R}^d} \mathscr{F}(\Delta_k f(x)) \mathscr{F}(\Delta_l g(x)) dx = \\ = \int_{\mathbb{R}^d} \delta_k \hat{f}(x) \delta_l \hat{g}(x) dx = 0$$

as $\delta_k$ and $\delta_l$ does not share support when $|k-l| > 1$. □

**Lemma 42.** *Let $f \in \mathcal{S}(\mathbb{R}^d)$. Then*

$$|| \sum_{k=-N}^{N} \Delta_k f - f||_2 \to 0 \text{ as } N \to \infty.$$

*In particular for any $\varphi \in \mathcal{S}(\mathbb{R}^d)$*

$$< \sum_{k=-N}^{N} \Delta_k f, \varphi > \to < f, \varphi > .$$

*Proof.*

$$|| \sum_{k=-N}^{N} \Delta_k f - f||_2^2 = \int_{\mathbb{R}^d} | \sum_{k=-N}^{N} \delta_k(\xi) \hat{f}(\xi) - \hat{f}(\xi)|^2 d\xi = \\ = \int_{\mathbb{R}^d} |\hat{f}(\xi)|^2 | \sum_{k=-N}^{N} \delta_k(\xi) - 1|^2 d\xi.$$

Now note that the sum in the last expression is zero for $2^{-N+1} < |\xi| < 2^N$ and in general bounded. Using this and the fact that $f$ lies in $L^2$ we get that in converges to zero. □

# Lecture 8

Now we let $f, g \in \mathcal{S}(\mathbb{R}^d)$ and look at $| < f, g > |$.

$$
\begin{aligned}
| < f, g > | &= | \lim_{N\to\infty} < \sum_{j=-N}^{N} \Delta_j f, g > | = \\
&= | \lim_{N\to\infty} \lim_{M\to\infty} < \sum_{j=-N}^{N} \Delta_j f, \sum_{k=-M}^{M} \Delta_k g > | \leq \\
&\leq \lim_{N\to\infty} \lim_{M\to\infty} \sum_{j=-N}^{N} \sum_{k=-M}^{M} | < \Delta_j f, \Delta_k g > | \leq \\
&\leq \sum_{j=-\infty}^{\infty} \sum_{l=-1}^{1} | < \Delta_j f, \Delta_{j+l} g > | \leq \\
&\leq \int_{\mathbb{R}^d} \sum_{j=-\infty}^{\infty} \sum_{l=-1}^{1} |\Delta_j f||\Delta_{j+l} g| dx \leq \\
&\leq \sum_{l=-1}^{1} \int_{\mathbb{R}^d} \left( \sum_{j=-\infty}^{\infty} |\Delta_j f|^2 \right)^{1/2} \left( \sum_{j=-\infty}^{\infty} |\Delta_j g|^2 \right)^{1/2} dx \leq \\
&\leq 3 \int_{\mathbb{R}^d} Sf(x) Sg(x) dx \leq 3 ||Sf||_p ||Sg||_{p'}
\end{aligned}
$$

Here at the last step we use Hölder and $\frac{1}{p} + \frac{1}{p'} = 1$. This gives us

$$
||Sf||_p \geq \sup_{g \in \mathcal{S}(\mathbb{R}^d),\ ||g||_{p'}=1} < Sf, Sg > \frac{1}{B_{p'}} \geq \frac{1}{3B_{p'}} \sup | < f, g > | = \frac{1}{3B_{p'}} ||f||_p
$$

and we are done. □

Recall the Calderon-Zygmund operator of convolution type theorem. In the proof we proved that the operator $T$ being $Tf = u * f$ for $f \in S(\mathbb{R}^d)$ was weak type $(1, 1)$. That is, it satisfied

$$
|\{x \in \mathbb{R}^d : |Tf| > a\}| \leq \frac{C}{a} ||f||_1
$$

ofr $f \in L^1(\mathbb{R}^d)$ This means that the distribution function of $Tf$ being $|\{x \in \mathbb{R}^d : |Tf| > a\}|$ multiplied by $a$ is bounded (precicely by $C||f||_1$). Functions like this are said to be in the weak $L^1$ space, denoted $L^{1,\infty}$. So for $f \in L^1$ we get that $Tf$ lies in $L^{1,\infty}$.
We will now try to construct a subset of $L^1$ such that $T$ of this subset lies in $L^1$.

**Definition 43.** Let $Q$ be a cube in $\mathbb{R}^d$ and let $a_Q$ be an $L^1$ function with support in $Q$ such that $\int_Q |a_Q|^2 dx \leq \frac{1}{|Q|}$ and $\int_Q a_Q dx = 0$. We call such a function $a_Q$ an $(L^2)$ atom on $Q$.

Recalling from the proof of the Calderon-Zygmund operator of convolution type theorem we see that $Ta_Q$ lies in $L^1$ if $a_Q$ is an atom on $Q$.

**Definition 44.** We define the space $X$ to be the set of functions $f \in L^1(\mathbb{R}^d)$ such that there exists $(\mu_k) \in \ell^1$ and sequence $(a_{Q_k})$ of atoms on $Q_k$ where

$$f = \sum_1^\infty \mu_k a_{Q_k}.$$

This is to be interpreted in the $L^1$ sense. On the space $X$ we have a norm

$$||f||_X := \inf_{f=\sum \mu_k a_{Q_k}} \sum_1^\infty |\mu_k|$$

and $X$ is a Banach space with this norm.

**Theorem 45.** *Let $T$ be an operator as in the Calderon-Zygmund operator of convolution type theorem. Then $Tf \in L^1$ for $f \in X$. and $T : X \to L^1$ is bounded.*

*Proof.* We know already that $||Ta_{Q_k}||_1 \leq C$ for all $k$, so $\sum_1^N \mu_k Ta_{Q_k}$ converges in $L^1$ to some $g$. We claim that $Tf = g$. In this case

$$||Tf||_1 \leq \sum_1^\infty \mu_k C \implies ||Tf||_1 \leq C||f||_X$$

To prove the equality let $\varepsilon > 0$. Then

$$|\{x \in \mathbb{R}^d : |(Tf)(x) - g(x)| > \varepsilon\}| =$$

$$= |\{x \in \mathbb{R}^d : |(Tf)(x) - (\sum_1^\infty \mu_k Ta_{Q_k})(x)| > \varepsilon\}| \leq$$

$$\leq |\{|(Tf)(x) - (\sum_1^N \mu_k Ta_{Q_k})(x)| > \frac{\varepsilon}{2}\}| + |\{|(Tf)(x) - (\sum_{N+1}^\infty \mu_k Ta_{Q_k})(x)| > \frac{\varepsilon}{2}\}| \leq$$

$$\leq \frac{C}{\varepsilon/2}||Tf - \sum_1^N \mu_k Ta_{Q_k}||_1 + |\{|(Tf)(x) - (\sum_{N+1}^\infty \mu_k Ta_{Q_k})(x)| > \frac{\varepsilon}{2}\}| \to$$

$\to 0$ as $N \to \infty$.

Thus $Tf = g$ almost everywhere. □

# Lecture 9

We constructed the space $X$ with a specific purpose so a natural question arises: Is the space $X$ optimal? or could we have made in bigger? It turns out that the answer to this question is a resounding yes, $X$ is optimal. To formulate this we start by defining the Riesz transform, the multidimensional generalization of the Hilbert transform.

**Definition 46.** The Riesz transform $R_j$ is defined by

$$R_j f := f * \frac{1}{\pi V_{d-1}} p.v.(\frac{x_j}{|x|^{d+1}}) \text{ for } f \in S(\mathbb{R}^d).$$

The Fourier transform of the Riesz transform $R_j$ is $-i\frac{\xi_j}{|\xi|}$. Now we also define the Hardy space.

**Definition 47.** The real hardy space $H^1(\mathbb{R}^d)$ is defined as

$$\{f \in L^1(\mathbb{R}^d) : f^* \in L^1(\mathbb{R}^d)\}$$

where

$$f^* = \sup_{|x-t|<y} |(f * P_t)(y)|.$$

Here $P_t$ is Poisson kernel

$$P_t(y) = \frac{\Gamma(\frac{d+1}{2})}{\pi^{\frac{d+1}{2}}} \frac{t}{(t^2+|y|^2)^{\frac{d+1}{2}}}.$$

$f^*$ is called the non-tangential maximal function of $f$.

**Theorem 48** (The Big Beautiful Theorem)**.** *Let $f \in L^1(\mathbb{R}^d)$, then the following are equivalent:*

$$\begin{aligned} &a) \quad f \in H^1(\mathbb{R}^d) \\ &b) \quad f \in X \\ &c) \quad R_j f \in L^1 \quad \forall j \in [1,d] \\ &d) \quad Sf \in L^1(\mathbb{R}^d) \end{aligned}$$

We will not prove this theorem as that is way above this course.
One question that arises is: what is the dual space of this space? The following result concerning this was originally proven in 1972 by Fefferman and Stine. They however, did not have access to the Big Beautiful theorem and the concept of atomic decomposition as in $X$ did not exist. Our proof here utilizing atomic decomposition is much simpler than their proof.

**Theorem 49** (Fefferman Duality Theorem)**.**

$$\left(H^1(\mathbb{R}^d)\right)^* = BMO(\mathbb{R}^d)$$

*The space $BMO(\mathbb{R}^d)$ is the space of bounded mean oscillation defined below.*

**Definition 50.** A function $f \in L^1(\mathbb{R}^d)$ is said to be of bounded mean oscillation if

$$||f||_{BMO} := \sup_{Q \subseteq \mathbb{R}^d} \frac{1}{|Q|} \int_Q |f(x) - < f >_Q |dx < \infty.$$

Here $Q$ is a cube and $< f >_Q := \frac{1}{|Q|} \int_Q f(x)dx$. Note that $|| \cdot ||_{BMO}$ is not a norm but a seminorm.